UNIVERSITY INSTITUTE OF COMPUTING

PROJECT REPORT

ON

STUDENT GRADE MANAGEMENT SYSTEM

Program Name: BCA Data Science
Subject Name/Code: C Programming (Code: _____)

Submitted by:	**Submitted to:**
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ABSTRACT

Introduction

The Student Grade Management System is a menu-driven C application designed to maintain basic student academic records. It accepts a student's roll number, name, and marks in five subjects. The program calculates the total marks and average, then assigns a grade according to a predefined grading scale. It also allows the user to display all stored records, search for an individual student using a roll number, and view the total number of students currently entered.

The project is intended to demonstrate how fundamental C programming concepts can be combined to solve a simple record-management problem. It provides a straightforward text-based interface and validates important inputs such as duplicate roll numbers and marks outside the permitted range.

Technique

1. Structures: A structure named *Student* groups related information, including roll number, name, subject marks, total, average, and grade.

2. Arrays: An array of structures stores up to 100 student records during one execution of the program. Each student record also contains an array for five subject marks.

3. Functions: Separate functions are used to add records, calculate grades, search records, and display student information. This makes the program easier to read and maintain.

4. Control structures and validation: Loops and switch-case selection control the menu. Conditional statements assign grades and check for invalid or duplicate entries.

System Configuration

Component	Requirement
Operating System	Windows, Linux, or macOS
Processor	Any modern processor capable of running a C compiler
Memory	4 GB RAM or higher recommended
Development Environment	Online C compiler (OnlineGDB) or any C IDE
Language / Compiler	C language; GCC or compatible C compiler

SUMMARY

Input

The application begins by displaying a main menu with five choices. The user can add a student record, display all student records, search for a student by roll number, view the total number of students, or exit the application.

When adding a student: the user enters a positive roll number, the student's name, and marks for five subjects. Each subject mark must be between 0 and 100. A roll number already present in the records is rejected.

When searching: the user enters the roll number of the student whose result is required. If a matching record exists, the full result is displayed; otherwise, a not-found message appears.

Process

After receiving the marks, the program adds the five values to calculate total marks. It divides the total by five to obtain the average. The average is compared with the grading scale and the corresponding grade is stored in the student record.

Average Marks	Assigned Grade
90 to 100	A+
80 to below 90	A
70 to below 80	B
60 to below 70	C
50 to below 60	D
Below 50	F

Output

The output is shown in the console. It includes confirmation after a successful record entry, a tabular list of all students, an individual result with subject-wise marks and calculated grade, the total number of students, and validation messages when the user enters an invalid or duplicate value.

PROGRAM FLOW

The following flowchart summarizes the main working sequence of the Student Grade Management System.

```
START
|
Display Main Menu
|
Read User Choice
|
+-- 1: Add Student
| |
| Validate Roll Number
| |
| Read Name and Five Marks
| |
| Calculate Total and Average
| |
| Assign Grade and Store Record
|
+-- 2: Display All Student Records
|
+-- 3: Search by Roll Number
| |
| Record Found?
| | Yes -> Display Result
| | No -> Show Not Found
|
+-- 4: Display Total Student Count
|
+-- 0: Exit
|
Return to Main Menu (unless Exit)
|
STOP
```

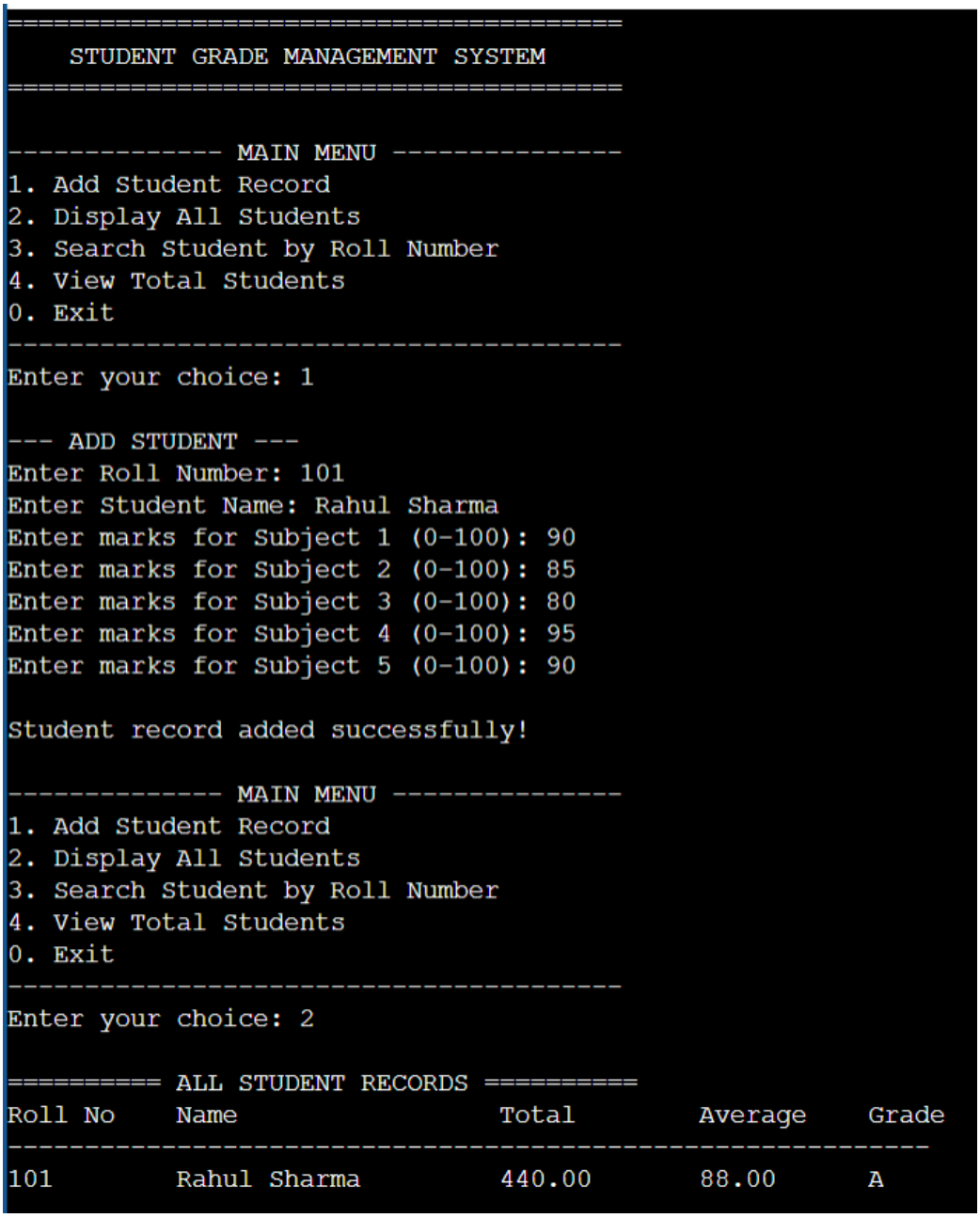
Limitations

The current version stores records in an array in the program's memory. Therefore, records are available only while the program is running and are cleared when it exits. The program supports up to 100 students and five subjects per student. File handling and permanent storage are not included in this version.

OUTPUT

The following screenshots were captured while testing the program in an online C compiler.

1. Main menu and successful student entry



```
========================================
    STUDENT GRADE MANAGEMENT SYSTEM
========================================

-------------- MAIN MENU ---------------
1. Add Student Record
2. Display All Students
3. Search Student by Roll Number
4. View Total Students
0. Exit
----------------------------------------
Enter your choice: 1

--- ADD STUDENT ---
Enter Roll Number: 101
Enter Student Name: Rahul Sharma
Enter marks for Subject 1 (0-100): 90
Enter marks for Subject 2 (0-100): 85
Enter marks for Subject 3 (0-100): 80
Enter marks for Subject 4 (0-100): 95
Enter marks for Subject 5 (0-100): 90

Student record added successfully!

-------------- MAIN MENU ---------------
1. Add Student Record
2. Display All Students
3. Search Student by Roll Number
4. View Total Students
0. Exit
----------------------------------------
Enter your choice: 2

========== ALL STUDENT RECORDS ==========
Roll No    Name                 Total        Average    Grade
------------------------------------------------------------
101        Rahul Sharma         440.00       88.00      A
```

2. Student result and total student count

```
-------------- MAIN MENU ---------------
1. Add Student Record
2. Display All Students
3. Search Student by Roll Number
4. View Total Students
0. Exit
---------------------------------------
Enter your choice: 3

--- SEARCH STUDENT ---
Enter Roll Number to Search: 101

---------------------------------------
          STUDENT RESULT
---------------------------------------
Roll Number : 101
Name        : Rahul Sharma
Subject 1   : 90.00 / 100
Subject 2   : 85.00 / 100
Subject 3   : 80.00 / 100
Subject 4   : 95.00 / 100
Subject 5   : 90.00 / 100
---------------------------------------
Total Marks : 440.00 / 500
Average     : 88.00%
Grade       : A
---------------------------------------

-------------- MAIN MENU ---------------
1. Add Student Record
2. Display All Students
3. Search Student by Roll Number
4. View Total Students
0. Exit
---------------------------------------
Enter your choice: 4

Total Students: 1
```

OUTPUT — VALIDATION TESTS

3. Duplicate roll number rejected and invalid marks handled

```
-------------- MAIN MENU ---------------
1. Add Student Record
2. Display All Students
3. Search Student by Roll Number
4. View Total Students
0. Exit
----------------------------------------
Enter your choice: 1

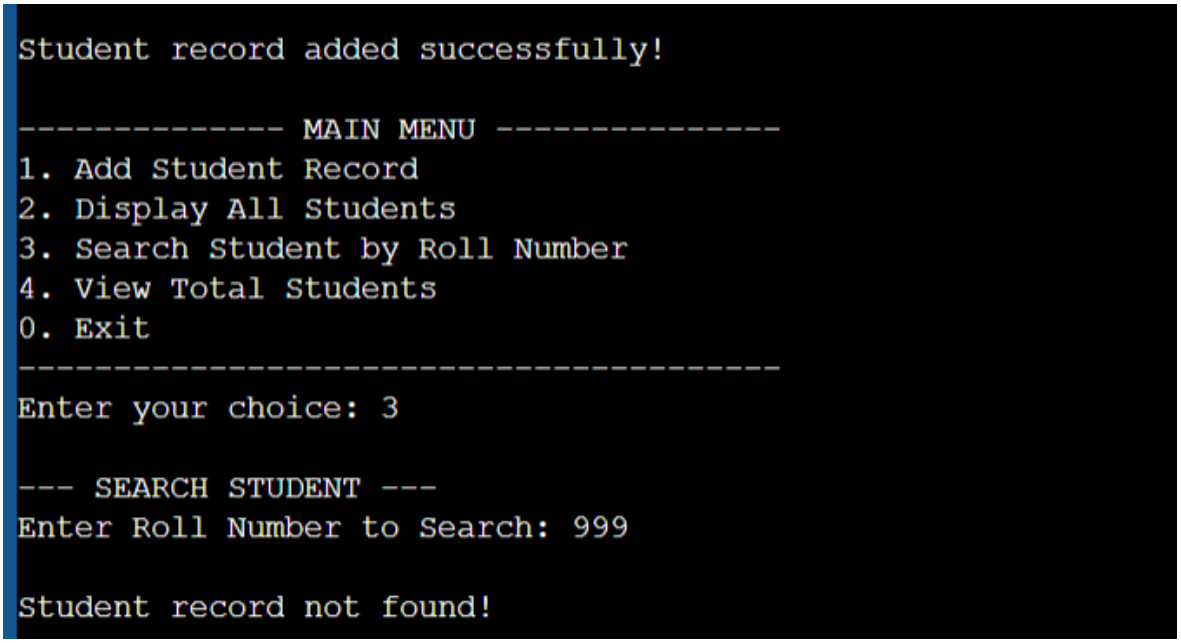
--- ADD STUDENT ---
Enter Roll Number: 101
This roll number already exists!

-------------- MAIN MENU ---------------
1. Add Student Record
2. Display All Students
3. Search Student by Roll Number
4. View Total Students
0. Exit
----------------------------------------
Enter your choice: 1

--- ADD STUDENT ---
Enter Roll Number: 102
Enter Student Name: Riya Metha
Enter marks for Subject 1 (0-100): 120
Invalid marks! Enter a value between 0 and 100.
Enter marks for Subject 1 (0-100): 90
Enter marks for Subject 2 (0-100): 95
Enter marks for Subject 3 (0-100): 80
Enter marks for Subject 4 (0-100): 85
Enter marks for Subject 5 (0-100): 87

Student record added successfully!
```

4. Search for a roll number that is not present



Conclusion

The Student Grade Management System demonstrates the use of structures, arrays, functions, loops, conditional statements, and input validation in C. The program successfully adds and displays student records, calculates totals and averages, assigns grades, and searches records by roll number. It provides a practical example of applying basic programming concepts to a simple academic record-management task.